Required One-Time FAST-LIO2 Installation

Husky GNSS-Denied Navigation — Team Instructions

Before running the Bayland command

FAST-LIO2 build products are excluded from Git. Cloning or pulling the repository supplies the source but does not create the executable. Perform this step on every new checkout or machine, from a fresh terminal.

```
cd <Journal_Work>/HIL/verified_stack
source /opt/ros/jazzy/setup.bash
bash hardware/build_fastlio.sh
source FASTLIO2Experiment/install/setup.bash
```

The build must finish without error. Do not start the experiment when `FASTLIO2Experiment/install/setup.bash` is missing.

Mandatory verification

```
test -x FASTLIO2Experiment/install/fast_lio/lib/fast_lio/fastlio_mapping \
  && echo "Correct FAST-LIO2 build exists"

ros2 pkg prefix fast_lio
```

The package prefix must begin with:

```
<Journal_Work>/HIL/verified_stack/FASTLIO2Experiment/install
```

If it resolves to `HIL/verified_stack/install`, another workspace, or an older installation, stop. Open a fresh terminal, source ROS Jazzy, rebuild with the command above, and repeat the verification. The same navigation command does not reproduce the same experiment when ROS selects a different FAST-LIO2 executable.

Then run the verified profile

Continue with Section 1 only after the local executable and package prefix have both been verified. Keep the complete result folder from every run. In `fast_lio2.log`, the launched executable must also be under `HIL/verified_stack/FASTLIO2Experiment/install/fast_lio/`.

Fail-safe rule: a missing local FAST-LIO2 build is a setup failure, not permission to fall back to an executable from another ROS overlay.

Husky HIL Main Instructions

FAST-LIO2 + SLAM Toolbox + Dijkstra Lattice

1 Gazebo Simulation

No hardware YAML changes are required for the packaged simulation.

This is the reference configuration for every later stage. It uses FAST-LIO2, SLAM Toolbox with scan matching and loop closing disabled, and Dijkstra Lattice with global exploration guidance, boundary following, straight-motion preference and an adaptive horizon with a 5 m base.

```
cd HIL/verified_stack

python3 localization_experiment/run_diagnostic.py \
  --map bayland \
  --robot a300 \
  --model DijkstraLattice \
  --localization slam \
  --slam-backend slam_toolbox \
  --pose-backend fast_lio2 \
  --explore \
  --boundary-follow \
  --straight-only \
  --adaptive-horizon \
  --experimental-horizon-m 5 \
  --no-slam-scan-matching \
  --no-slam-loop-closing \
  --no-record-bag
```

Remove `-no-record-bag` when recording is required. The verified stack supplies the Bayland world, A300 model, simulated LiDAR and IMU, FAST-LIO2 configuration, SLAM Toolbox, Dijkstra Lattice and mission targets.

Gazebo topics used by the verified stack

Purpose	ROS 2 topic	Gazebo topic
Clock	`/clock`	`/clock`
3D LiDAR	`/lidar/points`	`/husky/lidar3d/points`
IMU	`/imu/data`	`/model/husky/imu`
Wheel odometry	`/odom`	`/wheel/odom`
Velocity command	`/cmd_vel`	`/cmd_vel`
GNSS, evaluation only	`/gps/fix`	`/husky/gps`

2 Hardware in Loop

In this mode, the physical Husky computer runs localization and navigation. The physical Husky remains stationary. Gazebo supplies all sensor data, and only the simulated Husky moves.

Gazebo LiDAR + IMU → physical Husky computer
→ FAST-LIO2 + SLAM + Dijkstra Lattice
→ `/sim/husky/cmd_vel` → Gazebo Husky

Safety isolation

- Do not connect `/sim/husky/cmd_vel` to the Clearpath physical drive interface.
- Disable the physical motor driver or keep the physical E-stop engaged.
- Do not use physical LiDAR, IMU or wheel odometry in this test.
- Use the same `ROS_DOMAIN_ID` on the Gazebo and Husky computers.
- Verify bidirectional DDS communication before starting localization.

Isolated Gazebo command bridge

Change only the ROS side of the simulated command bridge:

```
- ros_topic_name: /sim/husky/cmd_vel
  gz_topic_name: /cmd_vel
  ros_type_name: geometry_msgs/msg/Twist
  gz_type_name: gz.msgs.Twist
  direction: ROS_TO_GZ
```

HIL robot profile

```
use_sim_time: true

localization:
  fastlio_config: fastlio.simulation.yaml
  pointcloud_topic: /lidar/points
  imu_topic: /imu/data
  calibration_confirmed: true

hardware_command_topic: /sim/husky/cmd_vel
```

Use the known simulated body-to-base calibration. Run navigation disarmed first and confirm the physical command interface receives no messages.

Run the same verified profile

From `HIL/`, start localization on the physical Husky computer:

```
python3 scripts/start_localization.py \
  --robot config/robot.hil.yaml
```

In another terminal, start the verified navigator disarmed:

```
python3 run_hil.py \
  --robot config/robot.hil.yaml \
  --mission config/mission.hil.yaml
```

After confirming that the physical drive interface remains isolated, set `verified_profile_armed: true` in `robot.hil.yaml` and repeat with `-arm`. In this HIL stage, arming permits commands only to `/sim/husky/cmd_vel`, so the simulated Husky moves. It must never be connected to the physical drive topic.

---

3 Physical Test

For a physical test, Gazebo is removed from the control loop. Create site-specific files:

```
cd HIL
cp config/robot.example.yaml config/robot.site.yaml
cp config/fastlio.velodyne.example.yaml config/fastlio.site.yaml
cp config/mission.example.yaml config/mission.site.yaml
```

Values to change in robot.site.yaml

```
base_frame: base_link
map_frame: map
odom_frame: odom_lidar
odom_topic: /fastlio/odometry_base
use_sim_time: false
verified_profile_armed: false

localization:
  calibration_confirmed: false
  fastlio_config: fastlio.site.yaml
  pointcloud_topic: <PHYSICAL_VELODYNE_TOPIC>
  imu_topic: <PHYSICAL_RAW_IMU_TOPIC>
  raw_odom_topic: /fastlio/odometry
  source_frame: camera_init
  source_body_frame: body
  body_to_base_translation: [x, y, z]
  body_to_base_quaternion: [x, y, z, w]

scan_topic: /scan
map_topic: /map
planner_command_topic: /hil/planner_cmd_vel
hardware_command_topic: <PHYSICAL_HUSKY_CMD_TOPIC>
```

Set `calibration_confirmed: true` only after verifying sensor clocks, frame directions and extrinsics.

Keep `verified_profile_armed: false` through sensor checks, localization startup and disarmed navigation. This value is the explicit gate for the later controlled physical-motion command.

Conservative first-test limits

```
max_linear_speed: 0.15
max_angular_speed: 0.20
linear_acceleration: 0.15
angular_acceleration: 0.25
reverse_speed: 0.08
```

Measure the physical footprint relative to `base_link`. Confirm that the physical command topic accepts `geometry_msgs/msg/Twist`; add a bridge if it expects `TwistStamped`.

Configure the physical Velodyne

The command below creates an editable site copy; it does not modify the example:

```
cp config/fastlio.velodyne.example.yaml config/fastlio.site.yaml
```

For a VLP-16 at 10 Hz with per-point time in seconds, begin with:

```
preprocess:
  lidar_type: 2
  scan_line: 16
  scan_rate: 10
  timestamp_unit: 0
  blind: 0.8
```

Timestamp code	Meaning
0	seconds
1	milliseconds
2	microseconds
3	nanoseconds

Enter the measured LiDAR-to-IMU transform:

```
mapping:
  extrinsic_T: [x, y, z]
  extrinsic_R: [1.0, 0.0, 0.0,
                0.0, 1.0, 0.0,
                0.0, 0.0, 1.0]
  extrinsic_est_en: false
```

The identity rotation is valid only when LiDAR and IMU axes point in exactly the same directions. Translation is in metres, with the LiDAR origin expressed in the IMU frame.

Required PointCloud2 fields

Each LiDAR point must normally contain `x`, `y`, `z`, `intensity`, `ring` and `time`. These are fields produced by the driver, not YAML values.

```
ros2 topic echo <velodyne-topic> --once --field fields
```

If the driver uses `t`, `timestamp`, or omits `ring` or `time`, stop and report the actual field description before running FAST-LIO2.

TF ownership

The required tree is:

```
map
 `-- odom_lidar
      `-- base_link
           |-- lidar_frame
           `-- imu_frame
```

Transform	Owner
`map -> odom_lidar`	SLAM Toolbox
`odom_lidar -> base_link`	supplied FAST-LIO adapter
`base_link -> lidar_frame`	robot URDF / robot_state_publisher
`base_link -> imu_frame`	robot URDF / robot_state_publisher

Do not disable `robot_state_publisher`. Disable or isolate only an existing localization component that publishes a competing dynamic parent for `base_link`.

```
ros2 topic info /tf --verbose
ros2 topic info /tf_static --verbose
ros2 run tf2_tools view_frames

ros2 topic echo <lidar-topic> --once --field header.frame_id
ros2 topic echo <imu-topic> --once --field header.frame_id

ros2 run tf2_ros tf2_echo base_link <lidar-frame>
ros2 run tf2_ros tf2_echo base_link <imu-frame>
```

There must be one publisher for each TF edge and one dynamic parent for `base_link`.

Mission coordinates

FAST-LIO2 initializes a local frame. The starting pose is normally near `(0,0)`; initial forward is positive X and left is positive Y.

```
targets:
  - {name: waypoint_1, x: 2.0, y: 0.0}
  - {name: goal, x: 4.0, y: 0.0}
```

Targets are metres in the active SLAM `map` frame, not latitude/longitude. Geographic targets require a surveyed start coordinate, initial bearing and conversion into local map X/Y before the mission.

Physical test sequence

Run from `HIL/` with the physical E-stop available.

The commands below use the same verified navigation profile as the reference simulation: FAST-LIO2, the verified SLAM policy, Dijkstra Lattice, global exploration guidance, boundary following, straight-motion preference and the adaptive 5 m base horizon. The site-specific topics, measured calibration, footprint, motion limits and mission coordinates are the only platform changes.

Terminal 1: sensor preflight

```
source /opt/ros/jazzy/setup.bash
source fastlio_ws/install/setup.bash

python3 scripts/preflight.py \
  --robot config/robot.site.yaml \
  --sensors-only
```

This checks LiDAR and IMU receipt, rates, timestamps, frame IDs, required point fields and mounting TF.

Terminal 1: start localization

```
python3 scripts/start_localization.py \
  --robot config/robot.site.yaml
```

Keep the Husky stationary while FAST-LIO2 initializes.

Terminal 2: full preflight

```
python3 scripts/preflight.py \
  --robot config/robot.site.yaml
```

This additionally checks odometry, scan, occupancy map and the current `map -> base_link` transform.

Terminal 2: disarmed navigation

```
python3 run_hil.py \
  --robot config/robot.site.yaml \
  --mission config/mission.site.yaml
```

Disarmed mode must output zero physical velocity. Verify target placement, route, map alignment, TF freshness and watchdog behavior.

After the disarmed checks pass, change only the following gate in `config/robot.site.yaml`:

```
verified_profile_armed: true
```

Terminal 2: armed controlled test

```
python3 run_hil.py \
  --robot config/robot.site.yaml \
  --mission config/mission.site.yaml \
  --arm
```

`-arm` does not select a different planner. It only permits the watchdog to forward bounded, fresh commands from `/hil/planner_cmd_vel` to the configured physical Husky command topic. The command is rejected while `verified_profile_armed` is false.

Information the integration team must return

- `ros2 bag info <bag-directory>` output.
- Exact LiDAR and raw IMU topics and message types.
- LiDAR PointCloud2 fields, datatypes and per-point time unit.
- LiDAR and IMU rates and frame IDs.
- Physical command topic and message type.
- `/tf` and `/tf_static` publisher information and generated `frames.pdf`.
- Measured LiDAR-to-IMU and body-to-base calibration.
- Robot footprint and tested motion limits.
- Site configuration, repository commit and ROS bag for each trial.

Acceptance conditions

Before counting a trial as successful, verify:

- GNSS and ground truth are absent from localization and planning inputs.
- All required messages and TF transforms are fresh and use one clock.
- The disarmed watchdog publishes zero velocity.
- Command loss, stale sensors or loss of localization readiness stops the robot.
- The physical E-stop operates independently of navigation software.
- Waypoint and goal arrival are measured independently from the reported estimate.
- Contacts, localization error, mission time and final physical goal error are recorded.